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**Role of Governments in ensuring transparency and accountability
of artificial intelligence systems in public administration**

Ensuring transparency and accountability of artificial intelligence systems in public administration**

Note by the Secretariat

The Secretariat has the honour to transmit to the Committee of Experts on Public Administration the paper prepared by Committee members Patricia Arriagada Villouta and Carlos Santiso, in collaboration with fellow Committee members Aigul Kosherbayeva, Sherifa Sherif and Najat Zarrouk.

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Ensuring transparency and accountability of artificial intelligence systems in public administration

Summary

The use of artificial intelligence in the public sector is growing with numerous potential benefits, such as enhanced operational efficiency, reduced cost and improved service delivery and decision-making. At the same time, its deployment can pose potential ethical, regulatory and operational risks and challenges that governments need to address.

In the present paper, the authors elaborate on the value of human oversight and regulation in ensuring that artificial intelligence systems are deployed strategically, responsibly and in an ethical manner, and that the use of these systems is conducted in a transparent and accountable manner and complies with international law. Examples of existing regulatory frameworks and policies are provided.

The authors also highlight the evolving role of supreme audit institutions in ensuring transparency and accountability of artificial intelligence systems in public administration as part of a broader accountability framework. This calls for empowering and equipping supreme audit institutions with the knowledge and information technology infrastructure required to use artificial intelligence systems themselves and to audit their use by government entities.

Examples are given of artificial intelligence applications in the public sector that have led to various socioeconomic benefits and improvements in public service delivery. The authors conclude with a set of recommendations for the Committee's consideration that could guide governments in the strategic, responsible and ethical deployment of artificial intelligence across public entities while safeguarding public interest.

I. Background

1. Artificial intelligence is increasingly recognized for its transformative potential in public administration, in particular for enhancing operational efficiency, cutting cost and improving decision-making. However, the adoption of artificial intelligence in public administration also poses ethical, regulatory and operational risks, as well as challenges that governments need to address.¹

2. Harnessing the potential of artificial intelligence to make governments work better and deliver better services is a critical challenge, but also a unique opportunity. Artificial intelligence is a game-changing technology that is poised to transform and disrupt public administrations around the world. While the global debate on the role of governments in the governance of artificial intelligence has tended to focus on their regulatory responsibilities to devise fit-for-purpose regulation, governments are also gearing up to using artificial intelligence themselves to reap its benefits.

3. In the broader context of the rapid evolution of artificial intelligence capabilities, making governments artificial intelligence-ready is crucial as it can make them more agile and responsive. As regulators, governments have a central role in steering and shaping the responsible development of artificial intelligence in the public interest. As users, they also need to ensure that they deploy artificial intelligence solutions within the public sector strategically, responsibly and in an ethical manner.

Artificial intelligence in the public sector today

4. A variety of artificial intelligence applications offer unique opportunities to improve the efficiency of government operations, cut costs and enhance the provision of public services to citizens.² This includes not only traditional artificial intelligence applications, but also the latest advancements in generative artificial intelligence. Artificial intelligence systems can handle large volumes of data, identify complex patterns and adapt and learn from new data to improve their accuracy.

5. Public entities are indeed critical users of artificial intelligence, reflected in a surge of cases in recent years. The use of artificial intelligence across core government functions is expanding rapidly, including in sensitive policy domains, such as taxation, law enforcement, justice administration and social welfare. Finland, for example, is using artificial intelligence to increase the uptake of applications for unemployment assistance and minimum income benefits. In Canada, the Revenue Agency has introduced artificial intelligence-powered chatbots to help people navigate complex information and procedures to facilitate compliance with tax obligations.

6. Other examples of artificial intelligence use in public administration concern property and infrastructure assessments through mapping technology, virtual assistants, facial recognition and artificial intelligence chatbots to increase citizen engagement.³ Artificial intelligence can be used across various stages of the policy

¹ Organisation for Economic Co-operation and Development, *Governing with artificial intelligence: are governments ready?* (OECD Publishing, Paris, 2024).

² Organisation for Economic Co-operation and Development and Corporacion Andina de fomento, *The Strategic and Responsible Use of Artificial Intelligence in the Public Sector of Latin America and the Caribbean* (OECD Publishing, Paris, 2022); Gavin Ugale and Cameron Hall, “Generative AI for anti-corruption and integrity in government: Taking stock of promise, perils and practice”, OECD Artificial Intelligence Papers, No. 12 (OECD Publishing, Paris, 2024).

³ Khuram Farooq and Bartosz Jakub Solowiej, *Artificial Intelligence in the Public Sector: Maximizing Opportunities, Managing Risks*, Equitable Growth, Finance and Institutions Insight (World Bank Group, Washington, D.C., 2020).

cycle from agenda setting, policy formulation and decision-making to implementation and evaluation. The Republic of Korea, for example, uses artificial intelligence to forecast emerging infectious disease and develop policy responses.

7. Governments are at different stages of development and implementation of national artificial intelligence strategies and policies. The development of national strategies and dedicated ministries focusing specifically on artificial intelligence is a relatively new phenomenon. The artificial intelligence strategy in the Dominican Republic, adopted in 2023, places a central emphasis on the responsible and strategic deployment of artificial intelligence in public administration. The national strategy for artificial intelligence in Norway emphasizes a coordinated approach to its deployment across public services, with dedicated oversight from agencies like the Norwegian Digitalisation Agency. In 2024, Spain established a Ministry for the Digital Transformation and Civil Service which is explicitly responsible for the steering and oversight of artificial intelligence deployment across the public sector.

8. While the potential benefits of artificial intelligence in the public sector are significant, its deployment has been fragmented and ad hoc with pilots that seldom scale for impact. This situation is, however, changing. In 2024, the Group of 20 mapped the development, deployment and adoption of artificial intelligence for enhanced public services, as part of a broader agenda for inclusive artificial intelligence. Similarly, the Group of Seven has started to address the specific challenges of artificial intelligence in the public sector and adopted a toolkit to guide deployment. The toolkit is a guide for policymakers and public employees to establishing actionable policies that help ensure that artificial intelligence is used in a safe, secure and trustworthy manner.

Benefits of artificial intelligence for the public sector

9. Enhancing effectiveness and improving efficiency. Artificial intelligence can lead to significant productivity gains across various government functions, streamlining processes and reducing costs, while also making government service delivery more efficient and effective. Repetitive physical and digital routine tasks can be automated, increasing operational efficiency, cutting payroll expenses and giving public servants the opportunity to concentrate on more important strategic responsibilities. Optimized processes can also lead to reduced resource waste and more efficient utilization of energy, resources and materials. Through data management and analytics, artificial intelligence can improve the targeting of public spending, enable more accurate and well-informed forecasting of demands and improve the management of resources, among others, ensuring equitable and efficient distribution.

10. Supporting evidence-based decision-making. Artificial intelligence makes it easier for public servants to make well-informed decisions by analysing vast amounts of data and uncovering trends and patterns, which may greatly improve the standard of services provided to the public.

11. Enhancing responsiveness. Artificial intelligence has the capacity to tailor and personalize public services to each citizen's unique requirements, thus enhancing their experience and satisfaction and allowing for proactive and integrated government services centred on the needs of people and their life events. Artificial intelligence can foster the engagement and participation of different stakeholders (e.g. through human-centric interfaces), which can help ensure that its use serves the public interest, while also identifying potential ethical concerns.

12. Ensuring compliance monitoring and risk management. Governments can use artificial intelligence for compliance monitoring and risk management, which has been proven effective in increasing revenue and detecting fraud. It can help raise red

flags in different areas, such as government procurement, tax obligations and welfare benefits. Some government auditors and procurement agencies have been particularly active in deploying artificial intelligence against corruption risks.⁴ To keep artificial intelligence risks within acceptable limits, governance mechanisms that match the magnitude of the risks are required.^{5,6}

13. Strengthening integrity, transparency and public trust. Artificial intelligence has become a powerful ally of transparency and oversight around the world and can assist in building integrity and public trust by being designed with explainable algorithms, actively addressing potential biases, providing clear justifications for decisions made and being used ethically, ultimately allowing stakeholders to understand how artificial intelligence systems work and trust their reliability.

14. Artificial intelligence can also be a potent force for accelerating the implementation of the 11 principles of effective governance for sustainable development, developed by the Committee and endorsed by the Economic and Social Council in July 2018, and its deployment in government needs to be guided by those very same principles.

Challenges and risks associated with the use of artificial intelligence in the public sector

15. The following section builds on the conclusions and recommendations of the Committee's twenty-third session. The use of artificial intelligence in the public sector poses ethical, regulatory and operational challenges and risks that need to be addressed in order to ensure that these systems are fair, inclusive, transparent and accountable, especially in developing countries with high levels of economic inequality and digital divides.⁷

16. Implementing artificial intelligence in the public sector remains a challenge in most countries owing to, inter alia, a lack of computing capacity; meaningful digital connectivity; data availability and accessibility; robust governance arrangements to steer digital government transformation from the centre of government; and digital skills and technical expertise.

17. Potential risks associated with artificial intelligence use include: further division between the privileged and the underprivileged; erosion of individual freedoms; personal privacy issues; security risks; discrimination and bias; lack of transparency and accountability; and the replacement of independent thought and judgment with automated control. For example, the European Union Artificial Intelligence Act, the legal framework to ensure that artificial intelligence systems deployed within the European Union are safe, trustworthy and respect fundamental rights, classifies most uses of artificial intelligence by governments as "high-risk" requiring heightened scrutiny, in particular in the areas of facial recognition, law enforcement and judicial administration.

18. Ensuring privacy, equity and non-discrimination emerges as a central concern, with many countries emphasizing the need for fairness and inclusiveness while

⁴ Carlos Santiso and Camilo Cetina, *DIGIntegridad: La transformación digital de la lucha contra la corrupción*. Corporación Andina de Fomento, 2022.

⁵ Yoshua Bengio and others, "Managing extreme AI risks amid rapid progress", *Science*, vol. 384, No. 6698, 2024.

⁶ Organisation for Economic Co-operation and Development, *Advancing accountability in AI: Governing and managing risks throughout the lifecycle for trustworthy AI*, *OECD Digital Economy Papers*, No. 349. Paris, 2023.

⁷ See E/2024/44-E/C.16/2024/9, chapter III.C.

undertaking steps to mitigate discrimination and exclusion. This is particularly important in the areas of social policies and welfare benefits.

19. Some country cases demonstrate the risks of discrimination and rights violations when artificial intelligence systems are not managed properly and are not integrated into legal and social systems with robust oversight. In one country, a childcare benefits scandal occurred where thousands of families were wrongfully accused of fraud by the tax authority based on flawed algorithmic risk assessments, resulting in the wrongful removal of more than one thousand children which were placed into foster care. In another country a failed attempt to recover money from alleged overpayments to welfare beneficiaries occurred, which led to issuing nearly half a million incorrect debt notices, putting many people under unwarranted financial stress.

20. Artificial intelligence-related challenges and risks also exist regarding the enforcement of criminal justice, with significant potential implications for human rights and fundamental freedoms. Artificial intelligence applications, such as predictive policing and extensive biometric data collection, can present a wide range of human rights issues.⁸ Algorithmic profiling systems often embed biases related to race, gender and socioeconomic status, thereby exacerbating inequalities. The use of data on nationality and ethnicity for risk-scoring needs to therefore be used with caution. For instance, predictive policing tools can reinforce biased data, thus leading to overpolicing in certain communities and perpetuating discriminatory attitudes and practices.

21. Artificial intelligence systems can also make judicial processes more complicated. Legal professionals and the accused may struggle to understand or challenge a decision that is algorithmically generated, thereby infringing on their right to a fair trial and due process. This is made worse by the inflexible nature of artificial intelligence, as fully automated decision-making lacks empathy, discretion and the ability to consider systemic disadvantages, resulting in rigid and potentially harsh outcomes. Justice administrators relying on artificial intelligence also run the risk of “automation bias”, which is humans’ tendency to take automated output without further scrutiny, thereby rendering human oversight ineffective and undermining the credibility of the justice system. In terms of data privacy, individual rights can be more easily violated by unlawfully gathered digital evidence, such as from hacked digital communication channels or facial recognition technologies, undermining the right to a fair trial. Digital evidence is also easier to forge and can increase the risk of wrongful convictions.

22. While there are critical risks to be mitigated especially in high-risk policy domains, such as social welfare, law enforcement and judicial administration, there also are many policy areas of lower risks where benefits can be significant.

II. The need for regulation and oversight

23. To address the challenges and risks associated with the use of artificial intelligence and, as further noted by the Committee at its twenty-third session, regulation, supervision and oversight through concerted global action are crucial to ensure responsible and ethical use of artificial intelligence, including in the public sector. Artificial intelligence needs to be developed and deployed by public administrations in a responsible, ethical and trustworthy manner for the benefit of all, especially the most vulnerable, mitigating risks of bias and discrimination, as well as

⁸ See report of the Secretary-General on human rights in the administration of justice (2024) (A/79/296).

exclusion of the so-called “data poor”. This includes establishing strong guardrails to ensure that artificial intelligence does not introduce new risks or inequalities nor exacerbates existing ones and does not violate human rights.

24. The development and deployment of artificial intelligence in the public sector, as well as its regulation and oversight, remain uneven between countries, as highlighted by the Government Artificial Intelligence Readiness Index published by Oxford Insights.⁹ While some countries are actively regulating the use of artificial intelligence in specific policy areas, such as healthcare or transport, few countries have established comprehensive mechanisms to evaluate and mitigate the specific risks and challenges that artificial intelligence may pose, with significant potential impacts especially for vulnerable groups. This gap underscores the need for enhanced attention and targeted strategies to ensure that artificial intelligence technologies not only advance public service efficiency but also promote social equity and inclusion.

25. Getting the governance of artificial intelligence right is a major challenge of digital government development. It is not about the technology itself, but how to best steer its use in the public interest and for common good, in line with our own principles, values and rights. Public administration also needs to always assess whether using artificial intelligence is indeed the best solution for a given problem by analysing alternative policy options, considering the trade-offs of using an automated approach and the associated costs for instance in terms of energy and environmental implications. An important dimension of the effective governance of artificial intelligence in the public sector also entails ensuring rules for appeal and means of redress for people to contest and correct infringements on their rights in the information society.

Issues of accountability

26. When artificial intelligence causes privacy violations or discrimination it is extremely complicated to determine accountability. The complexity of artificial intelligence systems often makes it particularly difficult to understand how decisions are made and with what degree of human control, while hindering the traceability of decisions, thus increasing the risk of bias, data privacy issues and infringement of human rights. The opacity of the so-called black box of algorithms and of the data on which they are trained and deployed on is a critical challenge for accountability as negative impacts cannot be identified without access to the inner workings and technical components of the system.

27. Consequently, the notion of algorithmic accountability has arisen recently, which is guided by fairness and integrity principles and aims to hold those who create and employ algorithms accountable for related outcomes, obliging them to make algorithms more transparent and putting efforts into mitigating associated risks.¹⁰ Policy instruments (e.g. white papers) and legal instruments, such as directives and bills, can be employed to enforce algorithm accountability. In fact, various countries are already implementing policies that require organizations to take responsibility for the impacts of their algorithms.

28. Algorithmic accountability requires a policy regime that combines audits, mandates for access to company data and impact assessments in order to address and mitigate risks. This regime is being rapidly mainstreamed across both the public and the private sector. The European Union Digital Services Act, for example, has several

⁹ Oxford Insights, *Government AI Readiness Index 2024*.

¹⁰ Pedro Cavalcante, *Artificial intelligence and algorithm accountability – potential risks and proposed governance measures*, Working paper, Blavatnik School of Government (University of Oxford, 2024).

clauses that mandate audits and give regulators and approved academics additional access to companies' data.

29. Similar efforts have been undertaken in the United States, such as the Generative Artificial Intelligence Accountability Act in California, which established oversight and accountability measures for the use of generative artificial intelligence within State agencies.

30. The Netherlands adopted the Fundamental Rights and Algorithms Impact Assessment to facilitate the assessment of risk at an early stage and prevent the premature use of an algorithm that has not been adequately assessed in terms of the consequences. Canada is using a mandatory Algorithmic Impact Assessment tool intended to support the Treasury Board's Directive on Automated Decision-making and determine the impact level of an automated decision-making system. Results are being made available to the public.

Ensuring transparency

31. The responsible and ethical deployment of artificial intelligence relies heavily on transparency, which encompasses several critical components:

- **Explainability and traceability:** information about the impact of automated decisions needs to be made available to users, allowing them to understand how and why decisions were made and where to turn for questions. This includes providing clarity regarding data inputs, methodologies used and the reasoning behind outcomes.
- **Data use and disclosure:** organizations need to be obligated to disclose what data is collected, how it is utilized and who has access to it. They also need to obtain consent for data processing from individuals. This is particularly vital for building consumer trust, especially considering widespread concerns about data privacy and potential misuse.
- **Algorithmic transparency:** organizations need to provide detailed information about their algorithms' training processes, including how they are assessed for biases and inaccuracies. This practice helps stakeholders understand the steps taken to ensure accuracy, fairness and reliability in artificial intelligence systems. Countries need to develop open registries of public sector algorithms as a first step towards greater transparency.
- **Communication:** maintaining an open and ongoing dialogue about the role, potential and limitations of artificial intelligence is essential for engaging stakeholders. This involves keeping users and the general public informed about changes to artificial intelligence systems (e.g. algorithmic adjustments or updates to data regulations) and potential inherent bias.

32. By prioritizing the elements listed above, governments can foster a more transparent and accountable artificial intelligence environment, ultimately enhancing public trust.

33. The Chilean Transparency Council, for example, has developed a binding General Instruction on Algorithmic Transparency for public entities. Other governments and cities (e.g. Amsterdam, Helsinki and New York) are establishing open registers of public algorithms. In France, the 2016 Digital Republic Bill provides a legal framework for algorithmic accountability and transparency requiring all public agencies to disclose information about their algorithms.

Existing regulatory frameworks and standards

34. International frameworks, norms and standards have been proliferating in recent years to guide the responsible, safe and trustworthy deployment of artificial intelligence across societies, economies and governments.

35. A number of principles to guide the governance of artificial intelligence exist which serve as a basis for the development of national and international standards. The Organization for Economic Co-operation and Development, for example, adopted the Recommendation on Artificial Intelligence in 2019 which outlines five guiding principles that also apply to the public sector: (a) inclusive growth, sustainable development and well-being; (b) human-centred values and fairness; (c) transparency and explainability; (d) robustness, security and safety; and (e) accountability.

36. The Recommendation on the Ethics of Artificial Intelligence, adopted in 2021 by the United Nations Educational, Scientific and Cultural Organization, outlines 10 core principles for a human-rights centred approach, namely: (a) proportionality and do no harm; (b) safety and security; (c) rights to privacy and data protection; (d) multi-stakeholder and adaptive governance and collaboration; (e) responsibility and accountability; (f) transparency and explainability; (g) human oversight and determination; (h) sustainability; (i) awareness and literacy; and (j) fairness and non-discrimination.

37. Going forward, there is a need to continue the ongoing global discussion and build a shared understanding of both the positive and negative impacts of artificial intelligence. Principles will have to be identified, policy priorities set and policy coherence ensured so that artificial intelligence can reinforce the 2030 Agenda for Sustainable Development and leave no one behind.

III. The role of supreme audit institutions as part of the broader accountability framework

38. Supreme audit institutions are part of the broader accountability framework and critical for enforcing transparency and accountability to prevent the misuse of artificial intelligence in the public sector. According to the International Organization of Supreme Audit Institutions, a supreme audit institution needs to be understood as a public body which performs the highest level of public sector auditing for a State or supranational organization in an independent and objective manner. The audit and jurisdictional responsibilities of supreme audit institutions vary based on their specific mandates and the legal frameworks in which they operate.

39. In general, the 195 supreme audit institutions gathered under the International Organization of Supreme Audit Institutions function as external auditors of governments, although there are exceptions (e.g. United States Government Accountability Office). Each government agency has internal audit bodies. In addition, a separate broader government audit entity exists. Some audit bodies are responsible for overseeing the proper use of public funds and ensuring that society's needs are adequately being met by government agencies while others concentrate on improving internal operations and controls of individual government agencies. External supreme audit institutions typically validate the work of internal controllers. All these entities are subject to the principles of transparency and accountability in the use of artificial intelligence.

40. The oversight performed by supreme audit institutions is a crucial part of the administrative process and needs to not be seen as separate or distant from it. Rather,

the administrative process needs to incorporate oversight and supervision as essential components. Oversight and administrative operations cannot function without one another, and imbalances in their capabilities need to be avoided. With regard to artificial intelligence, it is, therefore, essential that both supreme audit institutions and government agencies operate at similar levels of capacity, as imbalances could lead to insurmountable issues.

41. The International Organization of Supreme Audit Institutions has addressed artificial intelligence use by supreme audit institutions and the public sector in various ways, including through symposiums, the collection of experiences published in its International Journal of Government Auditing, and its Leveraging on Technological Advancement project. The latter is commendable as it supports supreme audit institutions in understanding the technological changes around them, assesses the implications of these changes for their work and empowers them to use and audit artificial intelligence systems. One of the key topics of the upcoming Congress of the International Organization of Supreme Audit Institutions in 2025 is the use of artificial intelligence techniques in auditing, but the organization already expressed interest in data analytics in 2016. Similarly, in 2019 its Congress encouraged supreme audit institutions to make better use of data analytics in audits.

The use of artificial intelligence by supreme audit institutions in performing their oversight role

42. When analysing what supreme audit institutions have done with regard to artificial intelligence, one can find varying levels of efforts and different approaches. Some have been particularly proactive in adopting artificial intelligence to better perform their oversight functions and responsibilities, while also taking an increasingly active role in ensuring effective oversight of artificial intelligence use across the public sector.

43. The Latin American and Caribbean Organization of Supreme Audit Institutions through its Committee on Information and Communications Technologies, for example, has conducted a regional survey which indicates that some supreme audit institutions are more advanced in the use and development of artificial intelligence systems (e.g. Brazil and Costa Rica), some are in the exploratory phase (e.g. Argentina, Chile and Colombia), and others have made no progress in the matter. Existing audit frameworks for algorithms and artificial intelligence accountability frameworks for government agencies and other entities referenced in this paper can be very helpful references for supreme audit institutions that are taking their first steps toward using and auditing artificial intelligence systems.

44. As a recent report of the World Bank pointed out,¹¹ there are wide differences between supreme audit institutions in the use of information technology-based tools and procedures. At one extreme, supreme audit institutions in some developing countries have little or no information technology facilities or infrastructure. At the other extreme, a small number of supreme audit institutions have the resources necessary to invest in the development, implementation, and use of highly sophisticated information technology-based audit tools and procedures. The majority of supreme audit institutions are somewhere in between and use software packages that are available commercially or under license, such as desktop analytics, electronic document management systems and specialist audit tools.

45. That said, it needs to be acknowledged that incorporating algorithmic systems into supreme audit institutions comes at a high cost, not only because of the required

¹¹ World Bank, *Supreme Audit Institutions' Use of Information Technology Globally for More Efficient and Effective Audits* (Washington, D.C., 2021).

equipment but also because the personnel using these systems needs to be adequately trained to understand the technical aspects, as well as social and ethical issues. Supreme audit institutions need to also not only understand artificial intelligence-based systems but also have been directly involved in their construction, implementation and development in case they are using them themselves. Highly skilled human talent is not easy to find and resources available to these oversight bodies are often insufficient for various reasons. This combination forms a challenging dynamic which warrants careful consideration as it could hinder the development of artificial intelligence within supreme audit institutions, both in terms of its use and oversight.

46. The high cost of implementation and investment in talent, combined with the fact that supreme audit institutions are largely dependent on the budgets allocated by the governments they oversee, create a risk of inadequate support. This issue warrants careful consideration.

47. Going forward, supreme audit institutions need to be encouraged to make a significant shift in how they conduct their audits, incorporating artificial intelligence into their methods. By preparing adequately, supreme audit institutions can ensure that artificial intelligence technologies are being used responsibly and effectively in public administration, thus safeguarding the public interest and ensuring transparency and accountability.¹² This requires increasing their budgets, seeking assistance from donors and collaborating with the community of supreme audit institutions which is better equipped to share its knowledge and experience in this field.

48. Innovative initiatives instrumental in testing artificial intelligence solutions for Supreme Audit Institutions worth mentioning are the innovation laboratories from the Supreme Audit Institutions of Belgium (DataLab), Brazil (coLAB-I), the Netherlands (Innovation Lab), Norway (Innovation Lab), the United States (Innovation Lab) and the European Court of Auditors (ECALab).

Ensuring the responsible use of artificial intelligence across the public sector

49. The International Organization of Supreme Audit Institutions has called upon its members to prepare for and leverage artificial intelligence. However, it has not issued standards or guidelines for auditing artificial intelligence-based systems, nor for ensuring transparency and accountability in the use of artificial intelligence by supreme audit institutions and government agencies.

50. If supreme audit institutions are not adequately equipped with the knowledge and information technology infrastructure required, it will be difficult for them to incorporate artificial intelligence-based systems into their tasks, let alone audit their use by government agencies. This can have wide reaching consequences as, for example, seen in the case of welfare scandals elaborated on in chapter I. The question then arises whether supreme audit institutions can issue guidelines related to the transparent and accountable use of artificial intelligence even if they themselves do not use artificial intelligence-based audit systems. Reviewing existing definitions of accountability and transparency can be very useful in this regard.

51. The Committee developed 11 principles of effective governance for sustainable development, endorsed by the Economic and Social Council in 2018. Three of these principles focus on accountability, namely: integrity (to serve in the public interest, civil servants are to discharge their official duties honestly, fairly and in a manner consistent with soundness of moral principle); transparency (to ensure accountability and enable public scrutiny, institutions are to be open and candid in the execution of

¹² Sutthi Suntharanurak, “Supreme audit institutions and preparing for the generative AI”, *Medium*, 2023.

their functions and promote access to information, subject only to the specific and limited exceptions as are provided by law); and independent oversight (to retain trust in government, oversight agencies are to act according to strictly professional considerations and apart from and unaffected by others).

52. The International Organization of Supreme Audit Institutions, for its part, has stated that accountability refers to the legal and reporting framework, organizational structure, strategy, procedures and actions to help ensure that:

- Supreme audit institutions meet their legal obligations regarding their audit mandate and required reporting within their budget;
- Supreme audit institutions evaluate and follow up on their own performance, as well as the impact of their audits;
- Supreme audit institutions report on the regularity and the efficiency of the use of public funds, including their own actions and activities and the use of Supreme Audit Institution resources;
- The head of the supreme audit institution, members of collegial institutions and the personnel of the supreme audit institution can be held responsible for their actions.

53. Through its principles on transparency and accountability, the International Organization of Supreme Audit Institutions has also clarified that transparency, which cannot easily be separated from accountability as both encompass many of the same actions, refers to the supreme audit institutions' timely, reliable, clear and relevant public reporting on its status, mandate, strategy, activities, financial management, operations and performance. Furthermore, it includes the obligation of public reporting on audit findings and conclusions and public access to information about the supreme audit institution.

54. Based on these definitions, supreme audit institutions have an obligation to oversee and monitor the use of artificial intelligence by public sector bodies, even if they do not have artificial intelligence-based applications themselves. Transparency could be ensured by clearly disclosing that artificial intelligence-based systems are being used, explaining their foundations in a simple manner, and publishing this information on public websites. For proactive transparency, it is recommended that organizations maintain publicly available information about artificial intelligence systems, particularly generative artificial intelligence, that could impact the fundamental rights of individuals or their access to services and benefits. Regarding accountability, supreme audit institutions can ensure the existence of proper governance, the adoption of best practices and clear, objective policies related to all procedures within the organization¹³ that are supported by artificial intelligence. All of this can be achieved through compliance audits.

55. However, this might not be enough, as the greatest challenge is to guarantee that artificial intelligence systems are fair and do not create or perpetuate biases. This includes checking the data sources, algorithms, and outcomes. In addition, it has to be assessed whether investments in artificial intelligence technologies are yielding the expected benefits and are aligned with public interest and whether artificial intelligence systems respect data privacy laws and prevent misuse of citizen data.

56. In this context, the United States Government Accountability Office has stated that accountability is driven by the tone at the top and supported by the commitment to integrity, ethical values, organizational structure and expectations of competence.¹⁴

¹³ Sydle, *Accountability: Qué es y cómo implementarla en las organizaciones*, 18 March 2024.

¹⁴ United States Government Accountability Office, *Standards for Internal Control in the Federal Government* (Washington, D.C., 2014).

Moreover, it notes that transparency is promoted by enabling external stakeholders to access information on the design, operation and limitations of artificial intelligence systems.¹⁵

National and regional approaches to artificial intelligence audit and oversight

57. The following examples illustrate that many supreme audit institutions are striving to stay up to date with artificial intelligence usage and development. However, on a global scale, efforts appear insufficient and much more work remains to be done. Nevertheless, sufficient interest, robust systems and promising initiatives exist, which will hopefully eventually lead to the correct approach for supreme audit institutions using and auditing artificial-intelligence-based systems

58. The Supreme Audit Institution of Brazil conducted a study in 2021 to understand the main artificial intelligence technologies used by the public administration, assess their current state and future prospects, and evaluate their role in control activities. Different artificial intelligence tools, such as ALICE (analysis of tenders, contracts and requests for tenders), ADELE (analysis of appeals on electronic tenders), MONICA (integrated monitoring of public procurement), and SOFIA (audit evidence guidance and analysis systems) have been used since 2015 for external public audits.¹⁶ For transparency, the Supreme Audit Institution discloses its use of artificial intelligence enabling public monitoring of the development and implementation of these technologies. The Supreme Audit Institution has also developed a guide for the use of artificial intelligence. While these initiatives show a commitment not to only adopt advanced technologies but to also ensure transparency, an accountability framework and dedicated related tools are currently missing.

59. In Chile, auditors have been using remote systems since 2011, both to audit services and to supervise their own auditors, through the Integrated Audit Control System. The country's Supreme Audit Institution is actively seeking out best practices and equipment to establish its own artificial intelligence-based audit system. In addition, a specialized unit on computing and artificial intelligence was recently created.

60. The Supreme Audit Institution of Costa Rica used artificial intelligence to audit public procurement conducted during the coronavirus disease (COVID-19) pandemic and subsequently developed a machine learning model to identify high risk contracts. It is also engaged in experimentation and proof-of-concept processes related to using artificial intelligence, including generating inputs for budget approval. Guidelines have been issued for the ethical and responsible use of artificial intelligence by the Comptroller General of the Republic, which include the mandatory publication of artificial intelligence applications use. To monitor the use of artificial intelligence-based systems by State agencies, the Supreme Audit Institution employs the "Public Management Monitoring: Management Capacity Index". While there is no specific oversight mechanism to ensure accountability or transparency, the Supreme Audit Institution is analysing the results of a 2024 survey of artificial intelligence use in public administration to incorporate specific oversight projects into its planning. Many State agencies are exploring artificial intelligence systems, but these are not yet operational, hence no direct audits of artificial intelligence use have been performed to date.

61. In Egypt, auditors are now required to acquire the necessary skills to utilize new technologies effectively. One major development is the embedding of artificial

¹⁵ United States Government Accountability Office, *Artificial Intelligence: An accountability framework for federal agencies and other entities* (30 June 2021) (GAO-21-519SP).

¹⁶ Teodora Lazăr Pleșa and others, "From Digitization to Artificial Intelligence in External Public Audit", *Valahian Journal of Economic Studies*, Volume 14(28), Issue 1, 2023.

intelligence into auditing, leading to the emergence of the concept of digital auditing. The Supreme Audit Institution of the Republic of Korea signed a Memorandum of Cooperation with the Institute of Artificial Intelligence at Seoul University in 2023, intending to develop artificial intelligence-based tools to be used for its external public audits. The Supreme Audit Institution of Nepal is preparing a predictive model based on issues identified in previous audits, such as tax miscalculations, to detect similar cases. The Supreme Audit Institution will use optical character recognition to automate the extraction of certain fixed field-based information from documents and will employ algorithms that reconcile revenue collection data to assist in drawing audit conclusions.

62. In the Netherlands, the Supreme Audit Institution has developed models and frameworks for different types of artificial intelligence audits, which consider ethical aspects of algorithms. An assessment framework for digital identity also exists. Together with Supreme Audit Institutions from Finland, Germany, Norway and the United Kingdom of Great Britain and Northern Ireland, the Supreme Audit Institution launched a white paper on auditing machine learning algorithms which aims to help supreme audit institutions and individual auditors by providing a set of guidelines and a methodology to perform audit tests.

63. The United States, through the Government Accountability Office, has been a pioneer of establishing regulatory frameworks for artificial intelligence. Its case is particularly interesting as its Artificial Intelligence Accountability Framework for Federal Agencies and Other Entities contains multiple recommendations for establishing artificial intelligence-based auditing systems. The framework distinguishes four principles for ensuring accountability: (a) governance (promote accountability by establishing processes to manage, operate and oversee its implementation); (b) data (ensure quality, reliability and representativeness of data sources and processing); (c) performance (produce results that are consistent with programme objectives); and (d) monitoring (ensure reliability and relevance over time). In this context, three broad approaches to human supervision of artificial intelligence are particularly relevant, namely: (i) human-in-the-loop (active human oversight of the artificial intelligence system, with the human retaining full control and the artificial intelligence only providing recommendations or input, which the human then reviews to make final decision); (ii) human-out-of-the-loop (lack of human supervision of the execution of decisions – artificial intelligence system has full control without the option of human override); and (iii) human-on-the-loop (human supervision – human has a monitoring or supervisory role, with the ability to take over control when the artificial intelligence model encounters unexpected or undesirable events). The authors suggest using the third approach, which is the continuous supervision of artificial intelligence systems by supreme audit institutions.

64. At a regional level, the European Court of Auditors developed an online methodology and guidance platform (AWARE) which helps guide auditors in order for them to select, plan, perform and report audits and non-audit engagements in the European Union in the most effective way. The Supreme Audit Institution of Estonia, through amendments to its organization law, allowed external public auditors access to large databases, which gave them the possibility to remotely analyse data from different fields or to interconnect different types of data, as well as to view contracts, invoices, the execution of budgets of public entities and much more. This contributed to the efficiency of audit planning and execution, especially in the case of financial audits.

IV. Selected examples of artificial intelligence use in the public sector

65. The following chapter illustrates selected case studies of artificial intelligence use in the public sector resulting in socioeconomic benefits and an improvement of public service delivery.

66. In recent years, Africa has witnessed a growing use of artificial intelligence in various sectors, aiming to address critical challenges, such as poverty, unemployment and misinformation. Some notable artificial intelligence initiatives across the continent are highlighted below, along with their impact on social welfare, employment and financial inclusion. In addition, several local and international artificial intelligence research hubs have been established. By continuing to leverage technology for social good, Africa can work towards breaking the cycles of poverty and unemployment and foster sustainable economic growth.

67. In Togo, for example, artificial intelligence has been effectively utilized to identify impoverished cantons, facilitating targeted allocation of social funds. In Zambia, artificial intelligence has been deployed to combat misinformation during elections, enhancing the integrity of the democratic process. These initiatives demonstrate the potential of artificial intelligence to drive positive social change, often supported by international partners. Artificial intelligence has also played a vital role in boosting financial inclusion across Africa. With two thirds of the continent's population lacking access to traditional financial services, many remain unbanked and excluded from economic opportunities. Mobile money solutions in Kenya exemplify how new technologies can address this issue, with the rise of mobile financial services enabling over 180,000 women to transition from farming to entrepreneurship, lifting them out of poverty and enhancing their income-generating potential.

68. In Colombia, the System of Identification of Social Programme Beneficiaries has been a cornerstone of the country's social welfare strategy since its inception in 1995. Designed to direct social spending towards the most vulnerable populations, the System employs an algorithmic approach to identify and classify citizens based on their socioeconomic status. This enables the Government to effectively allocate funds to anti-poverty programs, ensuring that assistance reaches those who need it most. In its fourth iteration, the System has integrated artificial intelligence and data analytics to identify inconsistencies within the System's database and detect potential fraud, allowing for more efficient and effective public spending.

69. In Egypt, as a result of the country's socioeconomic challenges, the Government has launched several successful social safety net programmes (e.g. Takaful and Karama, Hayat Kareema "The Decent Life Initiative"). These programmes provide conditional cash transfers, healthcare, housing and education. They also promote the financial inclusion of women in rural areas. With the support of the Egyptian Administrative Control Authority, the government has recently utilized artificial intelligence to enhance the efficiency of these programmes and ensure fair, equitable and accurate resource distribution across Egypt, reaching 10 million Egyptians across 27 governorates. The Administrative Control Authority also identified 44,000 undeserving recipients of the social safety programme Takaful and Karama only. This case demonstrates how the use of artificial intelligence can improve targeting and delivery of social safety net programs, by detecting and removing cases of fraud and ensuring that the most vulnerable benefit.

70. The COVID-19 pandemic accelerated Morocco's digital transformation through the creation of an enabling environment (e.g. institutional arrangements, collaborative governance at all levels, a national strategy deployed at the subnational level, mobilization of resources, capacity-building and awareness-raising programmes).

The country has undertaken major public sector reforms focusing on the management aspects of social protection systems, such as reforms of social security bodies, supervision of social protection systems, and the activation of the Single Social Registry which aligns with the Maroc Digital 2020 strategy, leading to the rapid adoption of advanced technologies, such as artificial intelligence, to streamline processes and improve service delivery. The Digital Development Agency played a crucial role in driving this transformation, funding artificial intelligence initiatives, establishing digital centres and supporting the integration of the youth into the digital economy. Going forward, the country's national digital strategy needs to adopt a more integrated vision and current digital laws need to be updated to better address the challenges and opportunities of artificial intelligence and foster its responsible development and deployment.

V. Recommendations

71. The authors welcome the Global Digital Compact, adopted by the General Assembly of the United Nations in September 2024, and confirm that to reap the benefits of artificial intelligence while mitigating its risks, the end goal for governments needs to be the strategic, holistic, ethical and responsible deployment of artificial intelligence in the public interest, steered from the centre of government, that addresses biases and ensures fairness, transparency and accountability. A “whole-of-government” and “whole-of-society” approach needs to be used to achieve the pledge of leaving no one behind and transform artificial intelligence challenges into opportunities. In this regard, the authors are making the following recommendations.

Need for regulation and oversight

72. Through concerted global action, governments need to focus on establishing the necessary structures, rules and procedures that direct the responsible and ethical use of artificial intelligence technology in the public interest.

73. In compliance with international law, transparency, accountability and robust human oversight of artificial intelligence systems need to be ensured. In addition, clear ownership and accountability structures within government departments for artificial intelligence initiatives need to be established.

74. Risk-based approaches to artificial intelligence deployment have been adopted by various governments to ensure that the benefits of artificial intelligence are harnessed in low risk, high-benefit policy domains. The establishment of a specialized regulatory body has been advanced by governments, in particular in the European Union, to assess and address potential risks and concerns about the wide use of artificial intelligence systems in decision-making.

Role of supreme audit institutions as part of the broader accountability framework

75. Supreme audit institutions play an important role as part of the broader accountability framework. Efforts by countries to empower and adequately equip them with the knowledge and information technology infrastructure required to use artificial intelligence systems themselves and to audit their use by government entities call for attention.

76. Regulators need to play an active role in shaping the audit landscape by providing guidance on best practices and establishing mandatory requirements for algorithm audits in high-risk sectors.

77. Developing open-source frameworks for algorithm auditing can facilitate collaboration and knowledge-sharing among auditors, researchers and practitioners, which could lead to more consistent practices across different organizations.

78. The International Organization of Supreme Audit Institutions is encouraged to: (a) establish clear guidelines for the use of artificial intelligence systems by its members; (b) assist less advanced supreme audit institutions in developing artificial intelligence applications, by seeking donor support, leveraging synergies among supreme audit institutions, and urging governments to provide the necessary budgets to ensure no one is left behind; and, (c) develop common approaches for auditing artificial intelligence use in public administrations.

Ensuring strategic and responsible deployment across the public sector

79. In terms of strategy, it is important to apply a whole-of-government approach. A number of approaches exist to steer and supervise the responsible deployment of artificial intelligence across the public sector, including:

- Creating robust governance structures and a supervision body at the centre of government in charge of the coordination of artificial intelligence deployment across the public sector in a consistent, strategic and responsible manner, as well as overseeing and enforcing artificial intelligence regulations. Such supervisory body could also train civil servants and coordinate with sector regulators on artificial intelligence-related challenges;
- Adopting specific, inclusive, accountable and human-centric policies and strategies that support national development objectives. These need to be linked to broader digital government strategies and/or connected with data governance strategies;
- Developing common guidelines that outline ethical standards, operational protocols and compliance mechanisms;
- Employing dynamic and continuously updated risk assessment protocols and customizing risk regulations based on the identified risks, while imposing stricter oversight for systems deemed higher risk. Artificial intelligence developers also need to be encouraged to proactively manage risk through regular reviews and self-assessment tools;
- Encouraging experimental approaches and testing of artificial intelligence applications in controlled environments or pilot projects to evaluate their reliability and societal impact, identify potential risks, refine algorithms and test fitness for purpose before wider deployment.

80. The strategic and responsible deployment of artificial intelligence in government would also be advanced by putting into place enablers to facilitate its uptake, such as:

- A solid and safe digital public infrastructure and, in particular, robust data infrastructure and governance (with data ethics considerations);
- A high degree and high quality of digital connectivity to ensure digital inclusion;
- A high level of maturity of cybersecurity systems;
- Adequate computing capacity and digital skills within government;

- Smart procurement rules and arrangements to procure artificial intelligence effectively and encourage innovation;¹⁷
- (Interdisciplinary) cooperation among State and non-State stakeholders and participation mechanisms to engage stakeholders in the development and evaluation process, resulting in enhanced transparency and accountability. Public consultations and participatory design approaches also allow affected communities to voice their concerns and contribute to the development of ethical guidelines;¹⁸
- Sustainable partnerships which can support trust-building and reciprocal learning. Novel partnerships with agile actors in the private sector, including government technology startups, can lead to co-creation of innovative digital solutions.

81. The responsible deployment of artificial intelligence in government also requires robust guardrails and approaches to mitigate bias and protect fundamental rights, including:

- Ensuring human oversight to ensure that artificial intelligence systems are ethical, transparent and accountable;
- Adopting ethical standards and mandatory procedures and embedding ethical and transparency considerations in the rules and protocols for government procurement of artificial intelligence;
- Adopting ex-ante due diligence and impact assessment frameworks and ex-post verification processes (including public participation, self-assessment and monitoring);
- Establishing clear accountability mechanisms which allow individuals and organizations to be held responsible for the outcomes of their artificial intelligence systems, paired with oversight procedures to monitor compliance with ethical standards;
- Developing protocols to allow for transparency, traceability and explicability of decisions reached using artificial intelligence. A human rights-centric approach needs to be applied, where governments actively inform individuals when they interact with artificial intelligence and where artificial intelligence decisions are broken down and explained;
- The design, data sources and decision-making processes of artificial intelligence systems need to be publicly disclosed in a comprehensible manner for non-technical users. Users also need to be allowed to challenge decisions made using algorithms, to provide feedback on the performance of these systems and have access to means of redress, including judicial review;
- Educating and raising awareness to address knowledge gaps and increase algorithmic literacy, while also supporting investigative journalism and oversight by civil society. This includes raising awareness among developers and users of algorithms regarding transparency and accountability.

¹⁷ World Economic Forum, AI Procurement in a Box Available at: <https://www.weforum.org/publications/ai-procurement-in-a-box/>.

¹⁸ Organisation for Economic Co-operation and Development, *Global trends in Government Innovation 2023. Trend 1: New forms of accountability for a new era of government*, OECD Public Governance Reviews (OECD Publishing, Paris, 2023).